Amir Mahdi Amani

Robotics & Computer Vision Engineer | AI Solutions

Location: Cologne, Germany

Bernhardstr. 119, 50968 Cologne

+49-152-11555803

amirmahdi.amani7th@gmail.com

Website: am-amani.github.io

GitHub: am-amani

Focus Areas

- Multi-Sensor Navigation
- Deep Reinforcement Learning
- Computer Vision
- Automation
- Geospatial Data Analysis

Skills

Programming:

Python	●●●●●
C++	●●●●○
SQL	●●●○○
MATLAB, R	●●●○○
Bash, LaTeX	●●●○○

AI and Vision:

PyTorch, TensorFlow	●●●●○
OpenCV, YOLOv5	●●●●○
Image Processing	●●●●○

Robotics:

ROS 1/2, Gazebo	●●●●●
SLAM, AMCL, MoveIt	●●●●○

Tools and Systems:

Linux, Docker, Git	●●●●○
QGIS, MS Office	●●●●○

Working Experience

Oct. 2024 – present
AI Solution Engineer – Network Planning, Optimization & Automation Specialist — **Deutsche Netzplanung GmbH, Cologne, Germany**
Contribute to the development and optimization of telecommunications infrastructure. Use QGIS to analyze surface areas and classify regions for efficient fiber-network deployment. Collaborate with senior engineers on technical solutions, analyze project data, and prepare detailed reports and stakeholder presentations. The role combines data analysis, software-supported planning, and technical communication.

Mar. 2024 – Sep. 2024
Fiber Network Planning and Optimization (Part-Time) — **Deutsche Netzplanung GmbH, Cologne, Germany**
Supported network-planning projects and the optimization of fiber infrastructure. Used QGIS for surface-area analysis and regional classification, reviewed technical data, assisted engineers with planning decisions, and supported implementation and project documentation.

Feb. 2024 – Sep. 2024
Robotic Research Assistant (HiWi) — **Humanoid Robots Lab, University of Bonn**
Analyzed and documented the Unitree GO1 architecture and operating protocols. Installed and configured ROS packages including SLAM and AMCL, tested different operating modes, prepared manuals for researchers, and upgraded TurtleBot2 systems for localization, navigation, and mapping to support ongoing laboratory work.

Sep. 2019 – Nov. 2019
Research Internship – Design-to-Robotic-Production and Operation — **TU Delft, The Netherlands**
Developed ROS-based solutions for research projects using MoveIt and Kinect V2. Applied forward and inverse kinematics for serial manipulators and implemented RGB-D object analysis through calibration, filtering, segmentation, clustering, object detection, and pose estimation. Gained practical exposure to the KUKA KR 10 and C4 Compact Controller and to industrial robotic-system workflows.

Education

Postgraduate Studies

Sep. 2021 – Sep. 2024
M.Sc. in Computer Engineering – Artificial Intelligence & Robotics
University of Padova, Italy
Thesis: Multi-Sensor Robotic Navigation Using a Teacher-Student Framework.
Grade: 96/110 (approximately 1.9/4).
Machine Learning | Computer Vision | Robotics | 3D Data

Dec. 2023 – Sep. 2024
Erasmus+ Exchange – Thesis and Research Internship — **University of Bonn, Germany**
Research at the Humanoid Robots Lab under Prof. Emanuele Menegatti, Prof. Maren Bennewitz, and Murad Dawood.
TD3 | LiDAR | Vision | ROS/Gazebo

Undergraduate Study

Sep. 2015 – Jun. 2020
B.Sc. in Computer Software Engineering — **University of Birjand, Iran**
Coursework in C++, data structures, algorithm design, robotics, and image processing. **Grade**: 14.93/20.

Other Training

2019 – 2021
Selected AI and Machine-Vision Certificates — **DeepLearning.AI / Stemmer Imaging**
TensorFlow for AI, machine learning and deep learning (2021); machine-vision solutions with HALCON and MERLIC (2019).

Short Bio

AI and robotics engineer with an M.Sc. in Computer Engineering, specializing in Artificial Intelligence and Robotics. Experienced in Python, C++, deep learning, computer vision, autonomous navigation, geospatial analysis, technical testing, and documentation. Research work includes multi-sensor robot navigation using teacher-student policy distillation in ROS/Gazebo. Currently developing data-driven solutions for network-planning and automation workflows in Germany.

Metrics

Profiles

Languages

English (Professional)

German (B1)

Personal

Interests include cooking, karaoke, dancing, reading, gym training, open discussions, and films. As a member of the Computer Science Association, helped organize an annual ACM programming competition for students from different universities in South Khorasan and conducted monthly C++ and Python workshops for freshmen. Also organized weekly English discussion sessions, special events, and Persian karaoke activities. Later created an online page for Persian karaoke content and community engagement.

Selected University Projects

Dec. 2023 — **Real-Time Camera Pose and Tumor Detection for Endoscopy** — Computer Vision
Combined camera-motion tracking with HSV filtering, Canny edge detection, Hough Circle Transform, background subtraction, and morphological operations for real-time pose estimation and potential-tumor detection.

Oct. 2023 — **TIAGO Mobile Robot Navigation and Obstacle Detection** — Robotics
Implemented ROS navigation from a user-defined start to a destination and developed an Action Client/Server system for detecting and reporting movable obstacles from laser-sensor data.

Jun. 2023 — **Image Colorization with Conditional WGAN** — Deep Learning
Built a PyTorch conditional WGAN with a U-Net architecture and skip connections, trained on more than 50,000 grayscale images, and used conditional inputs and image-processing techniques for controlled colorization.

May 2023 — **TinyPointNet for 3D Local Descriptors** — 3D Data Processing
Designed a compact PointNet variant for efficient local feature extraction from point clouds and evaluated it on diverse 3D datasets and benchmark tasks.

Jun. 2022 — **Human Hands Detection and Segmentation** — Computer Vision
Trained a custom YOLOv5 model with PyTorch and implemented C++ segmentation using contours, thresholding, and morphological operations.

Honours and Awards

Erasmus+ Traineeship Scholarship — University of Padova
Regione Veneto Scholarship (2.5 years) — University of Padova

Conference Experience

Mar. 2023 — **Speaker – 15th International Petroleum Technology Conference**
Bangkok, Thailand
Presented "Geological CO2 Leak Monitoring Using Joint Seismic-CSEM Inversion Based on CRIM and White Equations" on behalf of a peer and communicated the technical content to an international audience.

Teaching Experience

Teaching Assistant – Differential Equations — University of Birjand
Teaching Assistant – General Mathematics I — University of Birjand

References

Ref. 1 — **Prof. AmirHossein MajidiRad** — Purdue University Northwest
amajidir@pnw.edu

Ref. 2 — **Prof. Emanuele Menegatti** — University of Padova
emanuele.menegatti@unipd.it

Ref. 3 — **Prof. Maren Bennewitz** — University of Bonn
maren@cs.uni-bonn.de

Ref. 4 — **Prof. Alberto Pretto** — University of Padova
alberto.pretto@dei.unipd.it

Ref. 5 — **Prof. Nasim Nasrabadi** — University of Birjand
nasimnasrabadi@birjand.ac.ir

Publications

Journals

- Amani, A. M., Amani, S., and MajidiRad, A. H. "A Unified Conditional Policy for Multi-Robot Navigation via LiDAR-to-Vision Distillation." Accepted for publication in the Robotics, Mechatronics and Intelligent Machines section of *Machines* (MDPI), 2026.

Conference Presentation

- "Geological CO2 Leak Monitoring Using Joint Seismic-CSEM Inversion Based on CRIM and White Equations." Presented at the 15th International Petroleum Technology Conference in Bangkok on behalf of a peer, 2023.